核心文理-別冊解答

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目次

1　講義問題-解答

1.1　差分の基礎

1.1

▶解答◀

（1）$\Delta f(k)=k+1-k=\mathbf{1}$

（2）$\Delta f(k)=(k+1)^2-k^2=\boldsymbol{2k+1}$

（3）$\Delta f(k)=(k+1)^3-k^3=\boldsymbol{3k^2+3k+1}$

（4）$\Delta f(k)=(k+4)(k+3)(k+2)(k+1)-k(k+1)(k+2)(k+3)=\boldsymbol{4(k+1)(k+2)(k+3)}$

（5）$\Delta f(k)=2^{k+1}-2^k=\boldsymbol{2^k}$

（6）$\Delta f(k)=3^{k+1}\cdot(k+1)^2-3^k\cdot k^2=\boldsymbol{3^k\left(2k^2+6k+3\right)}$

（7）$\Delta f(k)=\dfrac{1}{(k+1)!}-\dfrac{1}{k!}=\boldsymbol{-\dfrac{k}{(k+1)!}}$

（8）$\Delta f(k)=\mathbf{0}$

（9）$\Delta f(k)=\mathbf{0}$

1.2

▶解答◀

（1）$A_n=\sum\limits_{k=1}^{n}1=\Big[\ k\ \Big]_1^{n+1}=\boldsymbol{n}$

（2）$B_n=\sum\limits_{k=1}^{n}k=\sum\limits_{k=1}^{n}\Delta\left(\dfrac{k(k-1)}{2}\right)=\left[\ \dfrac{k(k-1)}{2}\ \right]_1^{n+1}=\boldsymbol{\dfrac{n(n+1)}{2}}$

（3）$C_n=\sum\limits_{k=1}^{n}k^2$ を求める．

$k^2=k(k-1)+k$ と変形できるので，

$$
\begin{aligned}
C_n&=\sum_{k=1}^{n}\{k(k-1)+k\}\\
&=\left[\ \frac{(k-2)(k-1)k}{3}+\frac{k(k-1)}{2}\ \right]_1^{n+1}\\
&=\frac{1}{3}(n-1)n(n+1)+\frac{1}{2}n(n+1)\\
&=\frac{1}{6}n(n+1)(2n-2+3)\\
&=\boldsymbol{\frac{1}{6}n(n+1)(2n+1)}
\end{aligned}
$$

(4) $D_n = \sum_{k=1}^{n} k^3$を求める．

$k^3 = k(k-1)(k+1) + k$ と変形できるので，

$$
\begin{aligned}
D_n &= \sum_{k=1}^{n} \{k(k-1)(k+1) + k\} \\
&= \left[\frac{(k-2)(k-1)k(k+1)}{4} + \frac{(k-1)k}{2} \right]_1^{n+1} \\
&= \frac{(n-1)n(n+1)(n+2)}{4} + \frac{n(n+1)}{2} \\
&= \frac{1}{4}\{(n-1)n(n+1)(n+2) + 2n(n+1)\} \\
&= \frac{1}{4}\{n(n+1)(n^2+n-2+2)\} \\
&= \frac{1}{4}n(n+1)n(n+1) \\
&= \boldsymbol{\frac{1}{4}n^2(n+1)^2}
\end{aligned}
$$

(5)

$$
\begin{aligned}
E_n &= \sum_{k=0}^{n^2+1} k(k+1)(k+2)(k+3) \\
&= \left[\frac{1}{5}(k-1)k(k+1)(k+2)(k+3) \right]_0^{n^2+1} \\
&= \boldsymbol{\frac{1}{5}(n^2+1)(n^2+2)(n^2+3)(n^2+4)(n^2+5)}
\end{aligned}
$$

(6) $F_n = \sum_{k=-3}^{n-1} (k+2)(k+3)(k+4)(k+4)$ を求める．

まず，k の式を連続整数積の形に変形する．

$$
\begin{aligned}
(k+2)(k+3)(k+4)(k+4) &= (k+2)(k+3)(k+4)(k+1+3) \\
&= (k+1)(k+2)(k+3)(k+4) + 3(k+2)(k+3)(k+4)
\end{aligned}
$$

このことより，

$$
\begin{aligned}
F_n &= \left[\frac{1}{5}k(k+1)(k+2)(k+3)(k+4) + 3 \cdot \frac{1}{4}(k+1)(k+2)(k+3)(k+4) \right]_{-3}^{n-1} \\
&= \frac{1}{5}(n-1)n(n+1)(n+2)(n+3) + \frac{3}{4}n(n+1)(n+2)(n+3) \\
&= \text{つづき}
\end{aligned}
$$

(**7**)

$$
\begin{aligned}
G_n &= \sum_{k=1}^{n} 2^k \\
&= \left[\ \frac{2^k}{2-1}\ \right]_1^{n+1} \\
&= 2^{n+1} - 2 \\
&= \mathbf{2 \cdot (2^{n} - 1)}
\end{aligned}
$$

(**8**)$H_n = \sum_{k=1}^{n} (k+2) \cdot 3^k$ を求める.

今, $f(k) = (k+2) \cdot 3^k$, $F(k) = (pk^2 + qk + r) \cdot 3^k$　(p, r, q は定数)とおいて, $\Delta F(k) = f(k)$ を満たすように定数 p, q, r を定める.

$$
\begin{aligned}
\Delta F(k) &= \{\, p(k^2 + 2k + 1) + q(k+1) + r \} \cdot 3^{k+1} - (\, pk^2 + qk + r\,) \cdot 3^k \\
&= 3^k \cdot \{2pk^2 + 2(3p+q)k + 3p + 2q + 2r\}
\end{aligned}
$$

$f(k)$ とそれぞれ係数比較して,

$$
\begin{cases}
2p = 0 \\
6p + 2q = 1 \\
3p + 2q + 2r = 2
\end{cases}
\iff
\begin{cases}
p = 0 \\
q = \dfrac{1}{2} \\
r = \dfrac{1}{2}
\end{cases}
$$

$$
\therefore \quad F(k) = \frac{3^k}{2}(k+1)
$$

よって,

$$
\begin{aligned}
H_n &= \sum_{k=1}^{n} F(k) \\
&= \left[\ \frac{3^k}{2}(k+1)\ \right]_1^{n+1} \\
&= \boldsymbol{\left(\frac{n}{2} + 1\right) \cdot 3^{n+1} - 3}
\end{aligned}
$$

(**9**) $I_n = \sum_{k=1}^{n} k^4$ を求める．

まず，k^4 を連続整数積の形に変形する．

1.3

▶解答◀

（1）

$$
\begin{aligned}
\sum_{k=1}^{n}\{(k+1)^7-k^7\} &= \sum_{k=1}^{n}\Delta(k^7) \\
&= \Big[\,k^7\,\Big]_1^{n+1} \\
&= \boldsymbol{k^7-1}
\end{aligned}
$$

（2）

$$
\begin{aligned}
\sum_{k=1}^{n}\frac{\sqrt{k+1}+\sqrt{k}}{1} &= \sum_{k=1}^{n}(\sqrt{k+1}-\sqrt{k}) \\
&= \Big[\,\sqrt{k}\,\Big]_1^{n+1} \\
&= \boldsymbol{\sqrt{n+1}-1}
\end{aligned}
$$

（3）

$$
\begin{aligned}
\sum_{k=1}^{n}\frac{1}{k(k+1)} &= \sum_{k=1}^{n}\left(\frac{1}{k}-\frac{1}{k+1}\right) \quad (\text{☜部分分数分解}) \\
&= \left[\,\frac{1}{k}\,\right]_1^{n+1} \\
&= \boldsymbol{\frac{1}{n+1}-1}
\end{aligned}
$$

1.2　差分の応用

ほげ

1.3 漸化式

ほげ

1.4　二項定理

ほげ

1.5　場合の数の一番大切なこと

ほげ

1.6 種々の数え上げ

ほげ

2　各章の問題演習-解答

2.1　1章：差分の基礎

1.

▶解答◀

（**1**）

$$\begin{aligned}A_n &= \sum_{k=2}^{n}\frac{1}{k(K-1)} = \sum_{k=2}^{n}\left(\frac{1}{k-1}-\frac{1}{k}\right)\\ &= \left[\ \frac{1}{k-1}\ \right]_2^{n+1}\\ &= \boldsymbol{\frac{1}{n}-1}\end{aligned}$$

（**2**）

（**3**）

$$C_n = \sum_{k=1}^{n}k(k-1) = \left[\ \frac{1}{3}(k-1)k(k+1)\ \right]_1^{n+1} = \boldsymbol{\frac{1}{3}n(n+1)(n+2)}$$

（**4**）

$$\begin{aligned}D_n &= \sum_{k=-3}^{n}(k+2)(k+3)(k+4)\\ &= \sum_{k=-3}^{n}(k+2)(k+3)(k+4)\\ &= \left[\ \frac{1}{4}(k+1)(k+2)(k+3)(k+4)\ \right]_{-3}^{n+1}\\ &= \boldsymbol{\frac{1}{4}(n+2)(n+3)(n+4)(n+5)}\end{aligned}$$

（5）

$$\begin{aligned}E_n &= \sum_{k=0}^{n}(n-k)(n-(k-1))(n-(k-2))\\ &= \sum_{k=0}^{n}\Delta\left(\frac{1}{4}(n-(k-3))(n-(k-2))(n-(k-1))(n-k)\right)\\ &= \left[\ \frac{1}{4}(n-(k-3))(n-(k-2))(n-(k-1))(n-k)\ \right]_0^{n+1}\\ &= \frac{1}{4}\cdot 2\cdot 1\cdot 0\cdot 1-\left\{\frac{1}{4}(n+3)(n+2)(n+1)n\right\}\\ &= \boldsymbol{-\frac{1}{4}n(n+1)(n+2)(n+3)}\end{aligned}$$

2.

▶解答◀

（**1**）

$$A_n=\sum_{k=-n}^{n}3^k=\frac{3^n\cdot 2-3^{-n}}{3-1}\quad\left(☜\frac{（最後の次）-（最初）}{（公比）-1}\right)$$
$$=\frac{3^n}{2}\cdot\frac{8}{3}$$
$$=\boldsymbol{4\cdot 3^{n-1}}$$

（**2**）$B_n=\sum_{k=1}^{n}k\cdot 2^{-k}$

$f(k)=k\cdot 2^{-k}$, $F(k)=(pk^2+qk+r)$　（p, q, r は定数）とおいて，$\Delta F(k)=f(k)$ を満たすように

p, q, r を定める．

$$\Delta F(k)=2^{-(k+1)}\{p(k+1)^2+q(k+1)+r\}-2^{-k}\{pk^2+qk+r\}$$
$$=2^{-k}\left\{-\frac{1}{2}pk^2+\left(p-\frac{q}{2}\right)k+\frac{p}{2}+\frac{q}{2}-\frac{r}{2}\right\}$$

$f(k)$ とそれぞれ係数比較して，

$$\begin{cases}p=0\\-\dfrac{q}{2}=1\\-1-\dfrac{r}{2}=0\end{cases}\iff\begin{cases}p=0\\q=-2\\r=-2\end{cases}$$

$$\therefore\quad F(k)=-(k+1)\cdot 2^{-k+1}$$

$$B_n=\Big[-(k+1)\cdot 2^{-k+1}\Big]_1^{n+1}$$
$$=-\left(\frac{n+2}{2^n}-2\right)$$
$$=\boldsymbol{2-\frac{n+2}{2^n}}$$

(**3**) $\sum_{k=1}^{n} k^2 \cdot 3^k$ を求める．

$f(k) = k^2 \cdot 3^k$, $F(k) = (pk^3 + qk^2 + rk + s) \cdot 3^k$ （p, q, r, s は定数）とおいて，$\Delta F(k) = f(k)$ を満たすように p, q, r, s を定める．

$$\begin{aligned}\Delta F(k) &= (\,p(k+1)^3 + q(k+1)^2 + r(k+1) + s\,) \cdot 3^{k+1} - (\,pk^3 + qk^2 + rk + s\,) \cdot 3^k \\ &= 3^k \left[\,3\{\,p(k^3 + 2k^2 + 2k + 1) + q(k^2 + 2k + 1) + r(k+1) + s\} - (pk^3 + qk^2 + rk + s)\right]\end{aligned}$$

以下，[ ] の中だけ計算する．

$$\begin{aligned}[\ ] &= 2pk^3 + 6pk^2 + 5qk^2 + 6pk + 6qk + 2rk + 3p + 3q + 3r + 2s \\ &= 2pk^3 + (6p + 5q)k + (6p + 6q + 2r)k + 3p + 3q + 3r + 2s\end{aligned}$$

$f(k)$ それぞれ係数比較して，

$$\begin{cases} p = 0 \\ q = \dfrac{1}{5} \\ \dfrac{6}{5} + 2r = 0 \\ \dfrac{3}{5} + 3r + 2s = 0 \end{cases} \iff \begin{cases} p = 0 \\ q = \dfrac{1}{5} \\ r = -\dfrac{3}{5} \\ s = \dfrac{3}{5} \end{cases}$$

$$\therefore \quad F(k) = \left(\frac{1}{5}k^2 - \frac{3}{5}k + \frac{3}{5}\right) \cdot 3^k$$

$$\begin{aligned}\therefore \quad C_n &= \left[\ \frac{1}{5}k^2 - \frac{3}{5}k + \frac{3}{5} \cdot 3^k\ \right]_1^{n+1} \\ &= \frac{1}{5}\{\,((n+1)^2 - 3(n+1) + 3) \cdot 3^{n+1} - 3\} \\ &= \frac{1}{5}\{\,(n^2 - n + 1)3^{n+1} - 3\} \\ &= \boldsymbol{\frac{3}{5}\{\,(n^2 - n + 1) \cdot 3^n - 1\}}\end{aligned}$$